

GOOGLE CYBERSECURITY CERTIFICATE

DNS & ICMP Traffic Analysis

Google Cybersecurity Professional Certificate · Study Guide

Diagnosing a DNS outage from a tcpdump capture — recognizing an ICMP 'port unreachable' response pattern and narrowing the root cause down to either a downed DNS server or a firewall rule blocking port 53.

Track	Google Cybersecurity Professional Certificate
Course	Detection and Response
Tool	tcpdump
Analyst	Chayan Panchal

1 Problem Summary

Finding	Detail
Protocol — DNS layer	UDP was used to contact the DNS server for an A-record lookup of yummyrecipesforme.com
Protocol — error response	ICMP returned the error: udp port 53 unreachable
DNS query flags	The '+' after query ID 35084 marks flags on the UDP message; 'A?' requests an A record (domain -> IP mapping)
Root issue	Port 53 (DNS) is unreachable — the DNS server is not responding

NOTE Every log event shows the same shape: two lines of outbound UDP to the DNS server, followed by two lines of ICMP error response confirming DNS port 53 is blocked or down — a consistent pattern across the whole capture, not a one-off blip.

2 Incident Analysis

Time of incident	1:24 p.m. (13:24:32.192571 in the tcpdump log)
How reported	Customers contacted the company after receiving "destination port unreachable" errors trying to visit the website
Status	Under investigation by the security engineering team
Investigation steps	tcpdump packet sniffing performed — confirmed DNS port 53 unreachable
Next steps	Determine whether the DNS server itself is down, or a firewall rule is blocking port 53 traffic
Suspected root cause	Either a DoS attack took down the DNS server, or a firewall misconfiguration is blocking port 53

3 Quick Reference



DNS/ICMP triage — quick reference

DNS port 53 unreachable – investigation checklist:

1. Confirm the pattern: UDP query out, ICMP 'port unreachable' back, repeated
2. Check whether the DNS server process/host is actually up
3. Check firewall rules for port 53 (inbound and outbound)
4. Rule out a DoS against the DNS server as the underlying cause